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Metered Dose Topical Applicator

Abstract

A topical applicator of the propel/repel type includes an expandable reservoir with a port, an attachable applicator tip, and a metered dosing system. The reservoir can be a cylindrical housing with a movable elevator movable by a drive mechanism that is initially not engaged to allow filling of a flowable compound through the port, moving the elevator. An actuator, such as a rotating ring or pushbutton may engage the drive mechanism. The drive mechanism can include a one-way ratcheting feature. Metered doses can be dispensed through a hole or holes in the applicator tip by rotating a dosing knob or pushing a pushbutton causing the drive mechanism to advance the elevator. The drive mechanism and the action to dispense metered doses can be combined into a single motion. The drive mechanism can include detent positions that correspond to discrete doses, and can also provide audible and tactile feedback.

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Background/Summary

[0001] The present application is a continuation of U.S. application Ser. No. 17/610,139, filed Nov. 9, 2021, which is a National Phase entry of PCT Application No. PCT/US2020/032400, filed May 11, 2020, which claims priority to U.S. Provisional Application No. 62/845,789, filed May 9, 2019. Said applications are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The disclosed invention relates to devices and methods for preparing and dispensing pharmaceutical preparations, such as creams, liquids, and gels that are applied to the skin or other body surface. In particular, the disclosed invention relates to a metered dose topical applicator device suitable for filling with a flowable pharmaceutical compound, and that can be used to accurately dispense that compound and apply it to the body.

BACKGROUND OF THE INVENTION

[0003] One modality for administering therapeutic medicaments is by dermal application with subsequent transdermal absorption. This is quite common for hormone therapy, but is also used for pain medications and corticosteroids, among others. The therapeutic agent is typically blended into a carrier cream that is then rubbed onto the skin. Often, the formulation must be individualized for each consumer based on prescribed medicaments and dosages. These formulations are often prepared in compounding pharmacies, who then also prepare a container filled with the formulation and give it to the consumer. These containers are dispensers of various kinds, for example syringes or pump dispensers. Moreover, this modality is not limited to human adults, but can also be suitable for children, babies, and even pets such as cats and dogs.

[0004] Dispensing devices called propel/repel containers are common commercially available items used for cosmetics such as lip balm, lipstick, deodorants, antiperspirants, and moisturizers, and also for household and industrial products such as glue and lubricants. These have been adapted for topical application of therapeutic agents.

[0005] One of the most common configurations of propel/repel container features a hollow cylinder with a movable floor, female threads formed in the floor, a male-threaded rod passing through the female threads, and a screw knob, integral with the male-threaded rod, captured on one end of the cylinder. The product to be dispensed resides in the cylinder above the elevator. Turning the screw knob one direction forces the elevator upward, propelling the product from the cylinder, with the other direction repelling it. U.S. Pat. No. 1,499,784 to Becker discloses an example propel/repel container of this configuration, in this case for solid or semi-solid products like lipstick.

[0006] U.S. Pat. No. 3,616,970 to Baumann adds a closed exit end with an exit hole or holes, making it suitable for liquids and gels. U.S. Pat. No. 4,139,127 adds a ratchet mechanism to prevent repel motion, making it a propel dispenser only. U.S. Pat. No. 1,568,178 to Noble shows a configuration in which the drive screw is disposed entirely outside of the product reservoir, making the reservoir a simple, empty cylinder. U.S. Pat. No. 5,851,079 discloses a one-way ratchet mechanism as well as audible and tactile signals that are tied to metered incremental doses, called clicks, related to the volume of product dispensed.

[0007] The prior art also contains dispensers with drive systems that are engageable and disengageable, and that, further, feature the ability to be filled or refilled when the drive is

disengaged. An early example is the grease gun shown in U.S. Pat. No. 949,163 to Stapley. This device is, essentially, a screw thread driven syringe with a barrel with an outlet and a moveable plunger. It employs a lever that toggles on and off the engagement of a set of female threads to a threaded screw. Turning the screw expels grease from the outlet of the gun when the drive system is toggled on. When toggled off, the gun may be filled through the outlet because the plunger is not restrained by the thread engagement. Similar devices are disclosed in U.S. Pat. No. 2,283,915 to Cole, U.S. Pat. No. 3,353,718 to McLay, and U.S. Pat. No. 4,810,249 to Haber, et al, all of which could be used for, or are intended for, delivering medicaments.

[0008] U.S. Pat. No. 4,865,591 employs an engageable/disengageable drive combined with a metering system that is used to accurately dispense medicament from an attached medicament cartridge. It features a drive plunger with ratchet-style teeth that is engaged by mating toothed blocks to advance a plunger in the cartridge a distance preselected by the metering system. Disengagement of the drive plunger allows it to be quickly reset to a start position when replacing an empty cartridge with a new one.

[0009] Topical applicators for pharmaceutical formulations of liquids, creams, and gels with dose metering features also exist in the prior art. Examples of these devices are disclosed in U.S. Pat. Nos. 7,213,994 and 7,303,348 to Phipps, et al, and U.S. Pat. No. 8,544,684 to Perez. These are propel/repel containers of conventional construction with the additions of an indexed dose metering capability, ratchet mechanisms to prevent repel, and audible and tactile dose indicators. These are commercially available as the Topi-CLICK® from DoseLogix and the Ticker™ Transdermal Applicator from BIOSRX, respectively.

[0010] Another prior art metered dose topical applicator is disclosed in U.S. Pat. Pub. 2018/0207413 to Skakoon et al, titled “Metered Dose Topical Applicator,” which is commercially available as the UnoDose™ metered dose topical applicator from Reflex Medical.

[0011] All three of the commercially available applicators mentioned immediately above are best suited for adult patients to apply topical creams to large body surfaces such as a forearm. Even if they dispense small enough volumes per actuation (such as, for example, 50 microliters) their physical size is unsuitable for many applications, an example of which is applying a dispensed cream to the inside of pets' ears.

[0012] Thus, “micro-sized” metered dose topical applicators have been developed. Two example devices are the MD™ Pen and MD™ Syringe available from Medisca, Inc. of Montreal, Quebec, Canada. These devices feature an open medicament chamber that is filled with a therapeutic cream by a compounding pharmacy. After filling, a pump assembly with an outlet port is attached to the open end of the chamber. The pump is then actuated by pushing a pushbutton, on the opposite end of the chamber, in a motion similar to a clicking ball point pen. Each push of the pushbutton dispenses a metered dose of the cream out of the outlet port for dermal application. In this case, the dose size is 0.1 or 0.15 milliliters, depending on the model.

[0013] Two more examples of micro-sized metered dose topical applicators are the Topical TWIST Pen and the Topical CLICK pen available in the United States through RxCoop of Goldenrod, FL. These devices again feature an open medicament chamber that is filled with a therapeutic cream by a compounding pharmacy. After filling, an applicator tip assembly is attached to the open end of the medicament chamber. Both devices have a screw thread drive system that advances an elevator to expel the cream through a hole or holes in the applicator tip. In the Topical TWIST Pen version, rotating a knob on the opposite end of the medicament chamber advances the elevator. There are also a metering and anti-rotation functions included in the drive mechanism, which also include sensory feedback for the user. In the case of the Topical CLICK pen, pushing a pushbutton on the opposite end of the medicament chamber advances the elevator. This is similar to the action of a clicking ball point pen. Each click dispenses a metered dose and provides sensory feedback to the user. These devices dispense 0.01, 0.025, or 0.05 milliliters per metered dose, depending on the model.

[0014] The examples immediately above suffer drawbacks, the most obvious of which is inconvenient and ineffective filling. Because the medicament chambers are long and slender, the cream can only be added by using a long nozzle inserted into the chamber. Even then, filling the chamber without pockets of air is challenging. Moreover, removing any entrapped air bubbles can be almost impossible in some cases. This is especially detrimental to the overall performance and the accuracy of these micro-sized applicators because the metered-dose volumes are so small, and any bubbles can have a large negative effect.

[0015] Another example of a micro-sized metered dose applicator is disclosed in U.S. Pat. Application No. 2018/0178968 to Phipps, et al. A device based on this disclosure is commercially available as the Topi-Click MICRO™. This device may be filled with a compounded cream, the volume of which is discretionary between about 1 and 10 milliliters, and dispensed in 50 microliter increments (0.05 milliliters).

[0016] Fillable metered dose dispensers of the form disclosed in U.S. Pat. Application No. 2018/0178968 to Phipps, et al include: 1) a free-floating sealing plunger that moves during filling, and 2) a drive system that moves a plunger driver via a user action. The user action that moves the plunger driver first takes up the space, if any, between the plunger and plunger driver then moves the plunger as intended for dose dispensing. This arrangement, however, has disadvantages. In particular, if dispensers of this type are only filled partially, which is often the case, the distance between the plunger driver and the plunger can be large. This means that the dosing means (e.g. a rotating knob) must be actuated numerous times to take up that distance. Motorized tools (i.e. a battery-operated drill) can be employed, but that means having one available. Furthermore, stopping a motorized tool while never overshooting is difficult. Moreover, for the plunger driver to mechanically capture the plunger might be difficult or impossible, meaning that the plunger could possibly move forward during use independent of the plunger driver.

[0017] In light of the drawbacks of the prior art, there exists a need for an improved metered dose applicator that can be loaded with a flowable compound to any desired fraction of a full fill, and that can dispense the contents with a minimum of additional user handling steps.

SUMMARY OF THE INVENTION

[0018] Embodiments of the disclosed invention feature a device for incrementally metering discrete volumes of a compounded pharmaceutical liquid, cream, or gel formulations, and for topically applying the formulation for dermal absorption. The device is configured to allow any fraction of its maximum fill volume to be loaded when the drive mechanism is not engaged, yet allow the drive to be fully engaged immediately when desired by triggering an actuating element.

[0019] Embodiments can also include additional advantageous features as described in the following:

[0020] Embodiments can include a connection port through which compounded formulations may be introduced into an expandable and collapsible storage reservoir.

[0021] Embodiments can include a means of collapsing a storage reservoir, thus propelling the formulations out of the storage reservoir.

[0022] Embodiments can include an accurate means of propelling the formulation out of the device through the tip, such as an elevator driven by a drive mechanism comprising a screw thread and drive nut, which, in turn, may be actuated by user action such as rotating a knob.

[0023] Embodiments can include an actuating means that engages a drive mechanism upon demand.

[0024] Embodiments can include an attachable tip for the device with a plurality of openings to allow outflow of the formulation, and to provide a means of hands-free topical application of the formulation.

[0025] In embodiments, the nut can have flexible fingers with threaded portions at one end, a tubular end portion with cooperating ratchet portion at an opposing end. The cooperating ratchet portion may comprise a plurality of detent portions that engage recesses or openings in a surface of

the container housing. In embodiments, the recesses or openings may be positioned on a plate unitary with a barrel portion or sub-assembly of the container housing and may provide audible and/or tactile indication of incremental rotations corresponding to indexed metered doses.

[0026] In embodiments, the nut can have flexible fingers with threaded portions that are initially not engaged with screw threads of a drive mechanism, but that can be flexed via an operator actuation to become engaged.

[0027] Embodiments can include an indexing means that partitions the propelled formulation into metered doses.

[0028] Embodiments can include audible and tactile indications corresponding to the indexed metered doses.

[0029] Embodiments can include a one-way, or ratcheting, mechanism to prevent reversing of the propel action (repel).

[0030] Embodiments can include other useful features such as volumetric or other measurement scales, protective covers, and ergonomic geometric elements.

[0031] A further feature and advantage of the invention is that of a container housing configured as a barrel and elevator defining a reservoir that is conducive of holding a wide range of volumes of a compounded pharmaceutical liquid, cream, or gel formulation without needing a time consuming rotation of a knob to move the elevator forward, thereby bringing the liquid, cream, or gel formulation to the dispensing tip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] FIG. 1 is a perspective view of an embodiment of the metered dose topical applicator.

[0033] FIG. 2 is another perspective view of an embodiment of the metered dose topical applicator of FIG. 1 showing another configuration.

[0034] FIG. 3 is another perspective view of an embodiment of the metered dose topical applicator of FIG. 1 and FIG. 2 showing another configuration.

[0035] FIG. 4 is an exploded orthogonal view showing the components of the applicator individually.

[0036] FIG. 5 is a perspective, partially cut-away view showing the applicator in one configuration.

[0037] FIG. 6 is a detailed perspective view of one end of the applicator.

[0038] FIG. 7 is detailed perspective view, partially cut away, showing a middle portion of the applicator.

[0039] FIG. 8 is detailed perspective view, partially cut away, showing the other end of the applicator.

[0040] FIG. 9 detailed perspective view, partially cut away, showing the same end of the applicator as shown in FIG. 8, but viewed from a different angle.

[0041] FIG. 10 is a detailed perspective cross-section view showing components of the end of the applicator shown in FIG. 8 and FIG. 9.

[0042] FIG. 11 is a detailed perspective view, partially cut away, showing a middle portion of the applicator similar to FIG. 7, but in another configuration and from a different direction.

[0043] FIG. 12 is detailed perspective view, partially cut away, showing a middle portion of the applicator similar to FIG. 7, but in another configuration and with a component or components hidden.

[0044] FIG. 13 is a perspective view of one end of the applicator and with a filling syringe aligned to be attached to the applicator.

[0045] FIG. 14 is a perspective, partial cut-away view of a partially filled applicator that has dispensed a dose.

[0046] FIG. **15** is a perspective view of the applicator after filling.

[0047] FIGS. **16A** and **16B** are perspective views showing an alternate embodiment of a metered dose applicator.

[0048] FIGS. **17A** and **17B** are perspective views showing another alternate embodiment of a metered dose applicator.

[0049] FIG. **18** is a perspective view showing another alternate embodiment of a metered dose applicator.

[0050] FIGS. **19A** and **19B** are perspective views showing an alternate embodiment of a metered dose applicator.

[0051] FIG. **20** is a perspective view of an embodiment of a metered dose applicator with a pushbutton.

[0052] FIG. **21** is a cross-sectional view of the applicator of FIG. **20**.

[0053] FIG. **22** is a detailed cross-sectional view of the applicator of FIGS. **20** and **21**

[0054] FIG. **23** is a perspective view of the applicator of FIGS. **20** to **22**, with portions removed for clarity.

[0055] FIG. **24** is a detailed perspective view of the applicator of FIG. **23**, with portions removed.

[0056] FIG. **25** is a detailed cross-sectional view of a components of the applicator of FIG. **24**.

[0057] FIG. **26A** to **26C** are diagrammatic cross-sectional views of components of the applicator of FIGS. **20** to **25**.

[0058] FIGS. **27** and **28** is cross-sectional views of the applicator of FIGS. **20** to **26C**.

DETAILED DESCRIPTION OF THE INVENTION

[0059] Referring to FIG. **1**, an embodiment of a metered dose topical applicator **10** in accordance with the present invention is shown. The applicator **10** includes a barrel **100**, which has two regions, a medicament reservoir **102** and a mechanism housing **104**. The applicator **10** also includes a cover **1000**, a drive actuator **800**, and a dosing knob **600**, the functions of which are explained in detail below.

[0060] FIG. **2** shows the applicator **10** with cover **1000** removed to show an applicator tip **900**, which is used to topically apply cream that has been metered from the applicator **10**. The applicator tip **900** has an exit hole **902** through which the cream is dispensed from the applicator **10**.

[0061] FIG. **3** shows the applicator **10** in a configuration suitable for filling with a compounded cream. The applicator tip **900** has not yet been assembled, and the drive actuator has not been actuated, as shown by the notated gap shown in FIG. **3**. This configuration represents how applicator **10** may be provided to the user for preparation.

[0062] Referring now to FIG. **4**, internal components of one embodiment of applicator **10** are shown in an exploded view. Barrel **100** (with housing **104** and medicament reservoir **102**), applicator tip **900**, and cover **1000** have been exploded sideways for clarity, although in the actual assembly are coaxial with the other components. An elevator **200** sealingly and slidably rides in barrel **100** defining a volume in medicament reservoir **102**. Elevator **200** attaches to drive screw **300**. Drive nut **400** includes nut threads **404**, which are configured initially as disengaged to drive screw **300**, yet are engageable upon demand. Dosing knob **600** attaches to drive nut **400** via, for example, snap fit features. Retainer **500** assembles into housing **104** and provides capturing features that retain and position the rotatable subassembly of dosing knob **600** and drive nut **400**. An important function included in applicator **10** is its drive mechanism **12**, which comprises dosing knob **600** connected to drive nut **400**, which threadably and electively couples to drive screw **300** by way of the action of locking ring **700**. Briefly, rotating dosing knob **600** (as shown on FIG. **1**) turns drive nut **400**, which axially advances drive screw **300**. Drive screw **300** likewise pushes on elevator **200**, meaning that drive mechanism **12** expels cream out of applicator tip **900** of applicator **10**.

[0063] Referring now to FIG. **5**, elevator **200** is shown in the empty position within medicament reservoir **102** of barrel **100**, meaning that the compounded cream capacity of applicator **10** is

substantially zero. As shown in FIG. 6, elevator **200** includes a seal lip **202** that sealingly and slidably engages with the internal wall **112** of medicament reservoir **102**. In this embodiment, elevator **200** is separate from, but attaches to drive screw **300**, for example by a press or snap fit. Nevertheless, elevator **200** and drive screw **300** can be a single component without affecting the essence of the disclosed invention. Moreover, elevator **200** can be made from an elastomer, or can include an elastomer seal such as an O-ring, or can include a plurality of sealing lips, any of which could improve the seal integrity as required.

[0064] Referring to FIG. 5 and detail FIG. 7, drive screw **300** includes screw threads **302**, which can engage with nut threads **404** of drive nut **400**, which arrangement is shown here. Nut threads **404** are not yet engaged with screw threads **302** in FIG. 7, but become engaged during use of applicator **10**, as is further explained below.

[0065] Referring back to FIG. 5, nut threads **404** are integral with nut leg(s) **402**, which extends back to nut body **416**. Nut body **416** includes pawl(s) **412** and snapfit slot(s) **410**, which are also shown in detail FIG. 8. Retainer **500** is shown in its assembled position within housing **104** of barrel **100**. Retainer **500** may be affixed to housing **104** in various ways including a press or snap fit, adhesives, or welding. Retainer **500** includes retainer flange **506**.

[0066] Referring now to FIG. 9, dosing knob **600** is shown in its assembled position, in which snapfit leg(s) **602** have snapped into snapfit slot(s) **410** of nut body **416** in a typical snap-fit arrangement. Dosing knob **600** may be connected to drive nut **400** in various ways, the only functional criteria being that once the connection is made between drive nut **400** and dosing knob **600**, those two parts, once affixed together, are accurately and stably positioned within barrel **100**, yet are free to rotate about the main axis of applicator **10**.

[0067] Referring now to FIG. 10, retainer **500** includes ratchet tooth **502** and detent nest **504**, both of which work together with pawl **412** and detent bump **414**, which is integral with pawl **412**. Ratchet tooth **502** and pawl **412** allow rotation of drive nut **400** (and dosing knob **600**) in one direction, as shown by the arrow, yet check rotation in the opposite direction. Moreover, pawl **412** and ratchet tooth **502** can create an audible indicator of the passing of pawl **412** past ratchet tooth **502**, as is typical for ratchet and pawl mechanisms. Detent bump **414** and detent nest **504** work together to provide a stable detent position or positions at various rotational locations. This keeps the dosing knob **600** (and drive nut **400**) from rotating inadvertently while applicator **10** is not actively being used.

[0068] Referring back now to FIG. 5 and detail FIG. 7, nut leg **402** includes lock ramp **406** and lock nest **408**, whose purpose is to work together with locking ring **700** to effect engagement of nut threads **404** with screw threads **302**, as explained below. Drive actuator **800** is concentrically retained on the outer diameter of medicament reservoir **102**, yet is axially slidable between predefined stop positions. Drive actuator **800** includes actuator leg(s) **802** that contact locking ring face **704**. Thus pushing drive actuator **800** (to the right as shown in FIG. 7), will simultaneously push locking ring **700** to the right. Referring momentarily back to FIG. 1, barrel **100** includes a step feature, barrel stop **114**, between medicament reservoir **102** and housing **104**, which includes barrel slots **110** of barrel **100** through which actuator leg(s) **802** of drive actuator **800** fit. This combination acts to align and retain drive actuator **800** in its position.

[0069] Referring back to FIG. 7, as drive actuator **800** pushes locking ring **700** to the right, lock ramp **406** of nut leg **402** works together with guide slot **702** (see FIG. 12) of locking ring **700** to deform nut leg(s) **402** inward, which deformation engages nut threads **404** with screw threads **302**. Once locking ring **700** reaches its final position, it will be captured in lock nest **408** of nut leg **402**, and will rotate together with nut when dosing knob **600** (FIG. 5) is rotated.

[0070] Referring now to FIG. 11, drive actuator **800** has been pushed inward, or to the right, until it contacts barrel stop **114**, which action has also pushed locking ring **700** in the same direction, which has engaged nut threads **404** with screw threads **302**, and which has positioned locking ring **700** into a secure position in lock nest **408**. Nut leg **402** is essentially a cantilever spring, and has

been deformed inward by locking ring **700**.

[0071] Referring to FIG. **12**, locking ring **700** is shown in the locked position to where it has moved past lock ramp **406** into lock nest **408** (see FIG. **7**). (Drive actuator **800** has been hidden in FIG. **12** for a clearer view.) Guide slot **702** of locking ring **700** aligns and guides locking ring **700** along nut leg **402**. Because of the inward deflection of nut leg **402**, nut threads **404** are engaged with screw threads **302**, as shown. As previously noted, locking ring **700** rotates with drive nut **400** and dosing knob **600** (not shown in FIG. **12**).

[0072] FIG. **13** shows access port **106** of barrel **100**. Access port **106** includes port connector **108**, which can be, for example, a Luer style connector, in this case a male Luer lock. Any suitable connector style may be used, but a Luer configuration allows port connector **108** to attach directly to a Luer-tipped syringe (syringe **14** as shown). Syringe **14**, when connected, allows drug carriers such as liquids, creams, or gels to be added to medicament reservoir **102** of applicator **10** by pushing syringe plunger **16** forward.

[0073] We now turn to an explanation of the physical functionality of the present invention during use by referring to FIG. **5**. Applicator **10** may be supplied to a pharmacy for preparation in an empty state with elevator **200** substantially seated against end face **116** of barrel **100**. Thus, the volume held by medicament reservoir **102**, which is defined by the inside diameter of medicament reservoir **102** and the distance between end face **116** and elevator **200**, is substantially zero. Because nut threads **404** are not engaged with screw threads **302**, elevator **200** (and drive screw **300**) are free to move along the central axis of applicator **10**. Thus, for example, if syringe **14** has been used to inject a cream through access port **106** as shown in FIG. **13**, elevator **200** (and drive screw **300**) will move to accommodate a carrier cream **22** (not shown) by increasing the volume of medicament reservoir **102**.

[0074] Once the desired volume of a medicament and carrier cream **22** has been added to applicator **10**, drive actuator **800** is pushed in until it butts against barrel stop **114**, which, as explained above, engages nut threads **404** with screw threads **302**, which means the drive mechanism **12** (see FIG. **4**) of applicator **10** is likewise engaged.

[0075] Because the drive mechanism **12** is not engaged, i.e. nut threads **404** not meshed with screw threads **302**, during filling of applicator **10**, any desired percentage of fill volume, from 0 to 100% of medicament reservoir **102**, may be introduced. Engaging the drive mechanism **12** after filling, regardless of the fill volume, means the drive mechanism **12** is engaged only when desired, but immediately engaged when desired, with the single action of pushing drive actuator **800** to the position shown in FIG. **2**. Thus there is no need for a priming or take-up operation to accommodate medicament reservoir **102** fill volumes of less than 100%.

[0076] Actual dosing is shown in FIG. **14** wherein applicator **10** contains a quantity of carrier cream **22**, which can hold a suitable medicament. Drive actuator **800** has been pushed in, engaging nut threads **404** with screw threads **302** as described above. Dosing knob **600** has been turned in the direction shown so that a dose **24** of carrier cream **22** has been expelled out exit hole **902** of applicator tip **900**, and dose **24** is ready of application to the intended site.

[0077] Numerous other construction arrangements are possible that embody the functional characteristics of the present invention. For example, in FIG. **15**, a metered dose topical applicator **10** is shown in which its drive mechanism is also not engaged during filling, as described above. Applicator **10** also includes barrel **100**, dosing knob **600** and applicator tip **900**. In this alternative embodiment, however, the drive mechanism is engaged by rotating drive actuator **800** as shown, which engages nut threads and screw threads, which have been arranged internally to engage by this alternate action.

[0078] Likewise, FIG. **16A** shows an embodiment wherein metered dose topical applicator **10** includes side pushbutton **18**. After filling, side pushbutton **18** is pushed as shown by the arrow, which engages the drive mechanism in a fashion similar to what is described above. Side pushbutton **18** can latch, for example, in the pushed position to present a smooth outer surface to

barrel **100** as shown in FIG. **16B**.

[0079] An alternative embodiment is shown in FIG. **17A**, wherein applicator **10** has a drive mechanism that, as before, is not engaged to accommodate filling. Applicator **10** includes end pushbutton **20** that can be positioned proud of dosing knob **600** during filling, but when pushed as shown can trigger engagement of nut threads with screw threads as before. End pushbutton **20** can latch in a position flush with dosing knob **600** to present a smooth appearance, and to visually indicate engagement status as shown in FIG. **17B**.

[0080] Another alternative embodiment is shown in FIG. **18**. In this embodiment, the drive system is also initially not engaged to accommodate filling. Applicator **10** includes a dosing knob **26**, much like before, but in this embodiment, dosing knob **26** serves the dual function of also being the drive actuator. Thus, after the applicator has been filled with cream, dosing knob **26** is rotated as shown. This first causes the drive system's nut threads to engage with the screw threads, analogous to the internal action previously disclosed. Thereafter, rotating dosing knob **26** meters doses with sensory feedback and anti-rotation features as with other embodiments.

[0081] Another alternative embodiment is shown in FIG. **19A**. In this embodiment, the drive system is also initially not engaged to accommodate filling. In this embodiment, however, rather than a rotating knob, a pushbutton **28** is used to activate the metered doses. Moreover, pushbutton **28** can serve the dual function of also being the drive actuator. Thus, after the applicator has been filled with cream, pushing pushbutton **28** first causes the drive system's nut threads to engage with the screw threads, analogous to the internal action previously disclosed. That could, for example, put pushbutton **28** in the location shown in FIG. **19B**. Pushing further on pushbutton **28**, until it stops, then meters a dose by way of an internal drive mechanism similar to previously disclosed embodiments. In other words, pushing pushbutton **28** can rotate a drive nut, which then translates a screw, advances an elevator, and expels cream from the medicament reservoir. Releasing pushbutton **28** allows it to return to the position shown in FIG. **19B** by way of a return spring for example, which readies the applicator for another dose activation.

[0082] The embodiment shown in FIGS. **19A** and **19B** can be realized with various arrangements of functional internal components. One such arrangement is metered dose topical applicator **50** shown in FIG. **20**. Applicator **50** includes a barrel **52** and an applicator tip **54**. Drive housing **66** is assembled and affixed to barrel **52**. Pushbutton **74** is concentrically captured within drive housing **66**, but is free to move axially, within limits.

[0083] FIG. **21** shows the internal components of applicator **50**, starting with an elevator **56** and a screw **58**. In FIG. **21**, applicator **50** is shown in the configuration as supplied to the preparing agency (e.g. compounding pharmacy) in a state ready to be filled with medicament-containing cream. Elevator **56** and screw **58** can be a single component or an assembly of two affixed components, the preferred configuration defined only by manufacturing considerations. Disposed within barrel **52** is nut **62**, which will become engaged with screw **58** to drive elevator **56**, as will be explained below. Disposed within drive housing **66** is locking wedge **64** that, when repositioned, will effect the engagement of nut **62** with screw **58**. Also disposed within drive housing **66** is drive guide **68**, which is concentrically and axially affixed to barrel **52**. Disposed within drive guide **68** is drive actuator **70**, which is free to rotate and to translate axially, within limits. Pusher **72** is also disposed within drive guide **68**, and is free to translate axially, within limits. Pushbutton **74** is affixed to pusher **72**, and is the external link accessible to the user to initiate the internal drive system functioning of applicator **50**. Return spring **76** is disposed within drive housing **66**, and serves to bias drive actuator **70** to a start position, which bias can be overcome by the operator pushing on pushbutton **74**.

[0084] Referring now to FIG. **22**, nut **62** is shown within barrel **52** in the as-supplied configuration, i.e. ready to be filled. As with other embodiments of the present invention, nut **62** is not engaged with screw **58**, which allows screw **58** (and elevator **56**) to move axially during filling. Nut **62** includes two opposing flexible legs **78**, on which threaded portions **80** are disposed. As shown

threaded portions **80** are spaced away from screw **58**. Screw **58** includes screw threads **60**, but, because of the spacing, are not engaged with screw threads **60** in this configuration. Locking wedge **64**, shown here not in contact with nut **62** and legs **78**, serves to engage and lock the drive components, i.e. nut **62** with screw **58**, together for dispensing doses, as described below. Drive actuator **70** contacts locking wedge **64** in this configuration, as shown, and will urge locking wedge **64** to the left when applicator **50** is operated via pushbutton **74**.

[0085] Referring now to FIG. **23**, the assembled components of applicator **50** are shown in the same configuration as in the previous figure. Nut **62** is constrained axially, but is free to rotate about the axis, which rotation is accomplished with rotation of drive actuator **70**. Drive actuator **70** includes guide slots **82** that cooperate with drive lugs **84** of nut **62**, so that nut **62** rotates with drive actuator **70**. Nonetheless, drive actuator **70** is allowed to translate along the axis, within limits. Pusher **72** contacts drive actuator **70** as shown, and axial translation of pusher **72** to the left will translate drive actuator **70** as well. As previously described, pushbutton **74** is affixed to pusher **72**.

[0086] Referring now to FIG. **24** and FIG. **25**, the physical relationship between pusher **72** and drive guide **68** is shown. Pusher **72** includes guide lugs **86**. Drive guide **68** includes guide ribs **88**, which form guide rib slots **90** (FIG. **25**). Guide lugs **86** ride in guide rib slots **90** preventing pusher **72** from rotating during its allowed translation. Guide lugs **86**, guide ribs **88**, and pilot lugs **92** all include inclined planes **94**, which function as cams translating linear to rotary motion as will be explained further below.

[0087] Another important feature of applicator **50** is best described now by referring back to FIG. **23**. Just as guide lugs **86** of pusher **72** ride in guide rib slots **90** (FIG. **25**) of drive housing **66**, so, too, do pilot lugs **92** of drive actuator **70**, preventing drive actuator **70** from rotating, but in this case for only a portion of the allowable translation. Once pilot lugs **92** are no longer constrained by guide rib slots **90**, drive actuator **70** is free to rotate.

[0088] Having now defined the components of applicator **50**, the function of the drive system will be explained. Generally, pushing pushbutton **74** translates pusher **72**, which moves drive actuator **70** until it is no longer rotationally constrained. At that point, drive actuator **70** rotates by action of a spring force on inclined surfaces. This rotates nut **62**. Once pusher **72** reaches its travel limit, pushbutton **74** is released, and drive actuator **70** rotates again by additional action of the spring force supplied by return spring **76** on a different pair of inclined surfaces. This again rotates nut **62**. Rotation of nut **62** causes axial advancement of screw **58**, and elevator **56**, which do not rotate, to expel a dose.

[0089] This type of push-to-rotate mechanism is analogous, in part, to ordinary click-style ballpoint pens. In that case, the rotating member simply returns to either of two alternating positions, in or out, as the pushbutton is cycled. Adding female threads to the rotating member and a male-threaded screw, and eliminating the alternating return positions, are what make it into a dispensing mechanism rather than an alternating position mechanism. The aforementioned prior art device, the Topical CLICK pen available through RxCoop, uses a mechanism of this form.

[0090] Referring to FIGS. **26A**, **26B**, and **26C** shed additional light on the function of the drive mechanism of applicator **50**. For easier visualization, these figures show, conceptually, the key components “unwrapped” from their round configuration. Entities shown are block diagram analogs of those as described above, but with their entity numbers denoted by a prime ('). In the start position shown in FIG. **26A**, guide lug **86'** and pilot lug **92'** ride in guide rib slots **90'**. As previously explained pushing pushbutton **74** translates pusher **72**, and thus guide lug **86**, and, in turn, pilot lug **92**. Once pilot lug **92** moves beyond guide rib slot **90**, it rotates as shown in FIG. **26B** by action of the spring force on the mating inclined planes **94**. When pushbutton **74** is released, all components return toward the start position, rotated, however, to the next indexed location, as shown in FIG. **26C**.

[0091] Referring again to FIG. **22**, how the drive system of applicator **50** is changed from unengaged to engaged will be explained. Threaded portion **80** of nut **62** is not engaged with screw

threads **60** of screw **58** in the as-supplied, ready-to-fill configuration. As previously explained, this allows applicator **50** to be filled with any desired volume of a flowable compound. Once filled, the operator pushes the pushbutton **74** to dispense metered doses as previously described. However, the first actuation of pushbutton **74** is what causes threaded portion **80** to engage with screw threads **60**. This is shown in FIG. **27**, and in detail in FIG. **28**. Drive actuator **70**, when moved via pushbutton **74**, translates locking wedge **64** into a gap in nut **62** formed in legs **78**, which flexes legs **78** in such a way to force engagement of threaded portion **80** with screw threads **60**. Once in this position, locking wedge **64** remains there to keep the drive system engaged. FIG. **28** shows the now-engaged drive system in detail. Drive actuator **70** will be forced to return to its start position by return spring **76** once the operator releases pushbutton **74**.

[0092] So to summarize, applicator **50** has a drive system that is initially unengaged, which allows any desired amount of cream to be filled. Once filled, the first push of the dosing pushbutton permanently engages the drive system. Thereafter, each push dispenses a metered dose.

[0093] Additional permutations of actuating the drive system after filling, and dispensing metered doses after that, are possible. For example, one variation of the embodiment shown in

[0094] FIGS. **19A** and **19B** is to separate the drive actuation and dose activation functions into two different physical elements. Thus, this embodiment could include a side button as shown on FIG. **16A** to actuate the drive, leaving pushbutton **28** with the sole function of activating metered doses.

[0095] Other embodiments in accordance to the present invention may be constructed that present a drive actuation feature in any location and with any motion, being restrained only by whatever action is most convenient for the user, and which internal arrangement of components yields best to manufacturability. For example, no strict adherence to the elevator-to-drive screw-to-drive nut-to-dosing knob configuration is required to practice the present invention. The drive nut could be attached to, or integral with, the elevator, with the drive screw then attached to the knob. In other words, component locations and relationships can be inverted or otherwise changed and still function in the manner herein disclosed.

[0096] The following patents are incorporated by reference for all purposes: [0097] U.S. Pat. No. 949,163 to Stapley [0098] U.S. Pat. No. 1,568,178 to Noble [0099] U.S. Pat. No. 2,283,915 to Cole [0100] U.S. Pat. No. 3,353,718 to McLay [0101] U.S. Pat. No. 3,616,970 to Baumann et al. [0102] U.S. Pat. No. 4,139,127 to Gentile [0103] U.S. Pat. No. 4,810,249 to Haber, et al [0104] U.S. Pat. No. 5,725,133 to Iaia [0105] U.S. Pat. No. 5,851,079 to Horstman et al. [0106] U.S. Pat. No. 6,129,471 to Lang [0107] U.S. Pat. No. 6,551,611 to Elliesen et al. [0108] U.S. Pat. No. 7,086,564 to Corrigan [0109] U.S. Pat. No. 7,213,994 to Phipps et al. [0110] U.S. Pat. No. 7,303,348 to Phipps et al. [0111] U.S. Pat. No. 8,544,684 to Perez [0112] U.S. Pat. No. 8,950,993 to Gagne et al. [0113] U.S. Pat. No. 9,097,571 to Phipps et al. [0114] U.S. Pat. App. 2014/0031323 to Perez [0115] U.S. Pat. App. 2014/0221945 to Dos Santos et al. [0116] U.S. Pat. App. 2016/0129228 to Perez [0117] WO 2014/121259 to Dos Santos et al. [0118] U.S. Pat. App. No. 2018/0178968 to Phipps, et al. [0119] U.S. Pat. Pub. No. 2018/0207413 to Skakoon, et al.

[0120] The above disclosure is related to the detailed technical contents and inventive features thereof. People skilled in this field may proceed with a variety of modifications and replacements based on the disclosures and suggestions of the invention as described without departing from the characteristics thereof. Nevertheless, although such modifications and replacements are not fully disclosed in the above descriptions, they have substantially been covered by the spirit and technical theory of the subject invention.

[0121] While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and described in detail. It is understood, however, that the intention is not to limit the application to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure as defined by the appended claims.

[0122] Persons of ordinary skill in the relevant arts will recognize that various embodiments can

comprise fewer features than illustrated in any individual embodiment described above. The embodiments described herein are not meant to be an exhaustive presentation of the ways in which the various features may be combined. Accordingly, the embodiments are not mutually exclusive combinations of features; rather, the claims can comprise a combination of different individual features selected from different individual embodiments, as understood by persons of ordinary skill in the art.

[0123] References to “embodiment(s)”, “disclosure”, “present disclosure”, “embodiment(s) of the disclosure”, “disclosed embodiment(s)”, and the like contained herein refer to the specification (text, including the claims, and figures) of this patent application that are not admitted prior art. “Portion” when used herein may refer to a portion of a discrete component, all of a component, or a portion of an assembly, that is, for example, two components of an assembly of 5 components may be a “portion”, the terminology is not intended to be limiting.

[0124] For purposes of interpreting the claims, it is expressly intended that the provisions of 35 U.S.C. 112 (f) are not to be invoked unless the specific terms “means for” or “step for” are recited in the respective claim.

Claims

1. A topical applicator comprising: a barrel with a first end and a second end; an elevator slidably and sealably disposed within said barrel, the elevator attached or attachable to a threaded rod extending toward the first end of said barrel, the barrel and elevator defining a reservoir; an access port disposed on second end of said barrel through which flowable compounds may be added to or expelled from said reservoir; an applicator tip attachable to said access port, said applicator tip including a topical applicator surface and a hole or holes to convey said flowable compound from said reservoir to said applicator surface; a threaded nut with integral threads to reversibly engage threads of said threaded rod; a pushbutton disposed on first end of said barrel; an actuator disposed on or in relation to said barrel, wherein operating said actuating means toggles said threaded nut from disengaged to engaged with said threaded rod; wherein said flowable compound can be added to said reservoir, causing said elevator to translate in a first axial direction of said barrel, when actuating means has not engaged said threaded nut with said threaded rod, and, wherein pushing said pushbutton in a first direction translates said elevator in a second axial direction of said barrel when actuating means has engaged said threaded nut with said threaded rod to expel said flowable compounds through said access port to said applicator surface for topical application.
2. The topical applicator of claim 1, wherein pushing said pushbutton in a first direction operates said actuating means.
3. The topical applicator of claim 1, wherein the access port comprises one of a male luer and female luer connector and wherein the applicator tip includes the other of the male luer and female luer connector to sealingly attach to said access port.
4. The topical applicator of claim 1, wherein threads on said threaded nut are formed on fingers that flex toward said threaded rod to engage with threads of said threaded rod.
5. The topical applicator claim 1, wherein said actuator includes a ring disposed on said barrel that is translated in a first direction to effect engagement of said threaded nut to said threaded rod.
6. The topical applicator of claim 1, wherein said actuator includes a ring disposed on said barrel that is rotated in a first direction to effect engagement of said threaded nut to said threaded rod.
7. The topical applicator of claim 1, wherein said actuator includes a button disposed on said barrel that is pushed in a first direction to effect engagement of said threaded nut to said threaded rod.
8. The topical applicator of claim 1, including a ratchet to prevent rotation of said knob in a second direction.
9. A topical applicator comprising: a barrel with a first end and a second end; an elevator slidably and sealably disposed within said barrel, the elevator connecting to a threaded rod extending

toward the first end of said barrel, the barrel and elevator forming a reservoir; an access port disposed on second end of said barrel through which flowable compounds may be added to said reservoir; a threaded nut with a plurality of threaded portions movable radially inward from a disengaged position to an engaged position with the threaded rod; the threaded nut rotatable by and integral with a knob; an actuator disposed on or in relation to said barrel, wherein operating said actuator moves the threaded portions of the nut radially inward from the disengaged position to the engaged position with said threaded rod; wherein said flowable compound can be added to said reservoir at the access port, causing said elevator to translate in a first axial direction of said barrel when the threaded portions of the nut are not engaged with said threaded rod, and, wherein rotating said knob in a first direction translates said elevator in a second axial direction of said barrel when the plurality of threaded portions of said threaded nut are engaged with said threaded rod to expel said flowable compounds from said second end of said applicator for topical application.

10. The topical applicator of claim 9 further comprising an applicator tip attachable to said access port, said applicator tip including a topical applicator surface and a hole or holes to convey said flowable compound from said reservoir to said applicator surface, wherein the access port comprises one of a male luer or female luer connector.

11. The topical applicator of claim 10, wherein threads on said threaded portions of the nut are formed on fingers that flex toward said threaded rod to engage with threads of said threaded rod.

12. The topical applicator of claim 9, wherein said actuator includes a ring disposed on said barrel with cam surfaces that is translated in a first direction to effect deflection of the threaded portions toward said threaded rod.

13. The topical applicator of claim 9, wherein said actuator includes a ring disposed on said barrel that is rotated in a first direction to effect engagement of said threaded nut to said threaded rod.

14. The topical applicator of claim 9, wherein said actuator includes a button disposed on said barrel that is pushed in a first direction to effect engagement of said threaded nut to said threaded rod.

15. The topical applicator of claim 9, wherein said actuator includes a latch in physical communication with said ring or button, said latch maintaining engagement of said threaded nut to said threaded rod.

16. The topical applicator of claim 15, wherein said latch is a ring slidably or rotatably disposed on said nut, said ring causing said fingers to flex, thereby engaging said threaded nut to said threaded rod.

17. The topical applicator of claim 16 wherein said ring cooperates with latching features on said fingers such that threaded nut remains engaged with threaded rod.

18. The topical applicator of claim 9, further including a ratchet to prevent rotation of said knob in a second direction.

19. A method of filling and using a topical applicator, comprising: injecting a flowable compound into a reservoir defined by a barrel and an elevator in the barrel in an elevated position in the barrel, the barrel connected to a manual operated drive mechanism comprising a nut and a cooperating screw, the nut and screw is a non engaged position, whereby the injection of the flowable compound moves the elevator downwardly and causes relative motion between the nut and screw; engaging the nut and the cooperating screw by an externally exposed actuator on the topical applicator.

20. The method of claim 19, further comprising dispensing the flowable compound by rotation of a knob connected to the drive mechanism.
